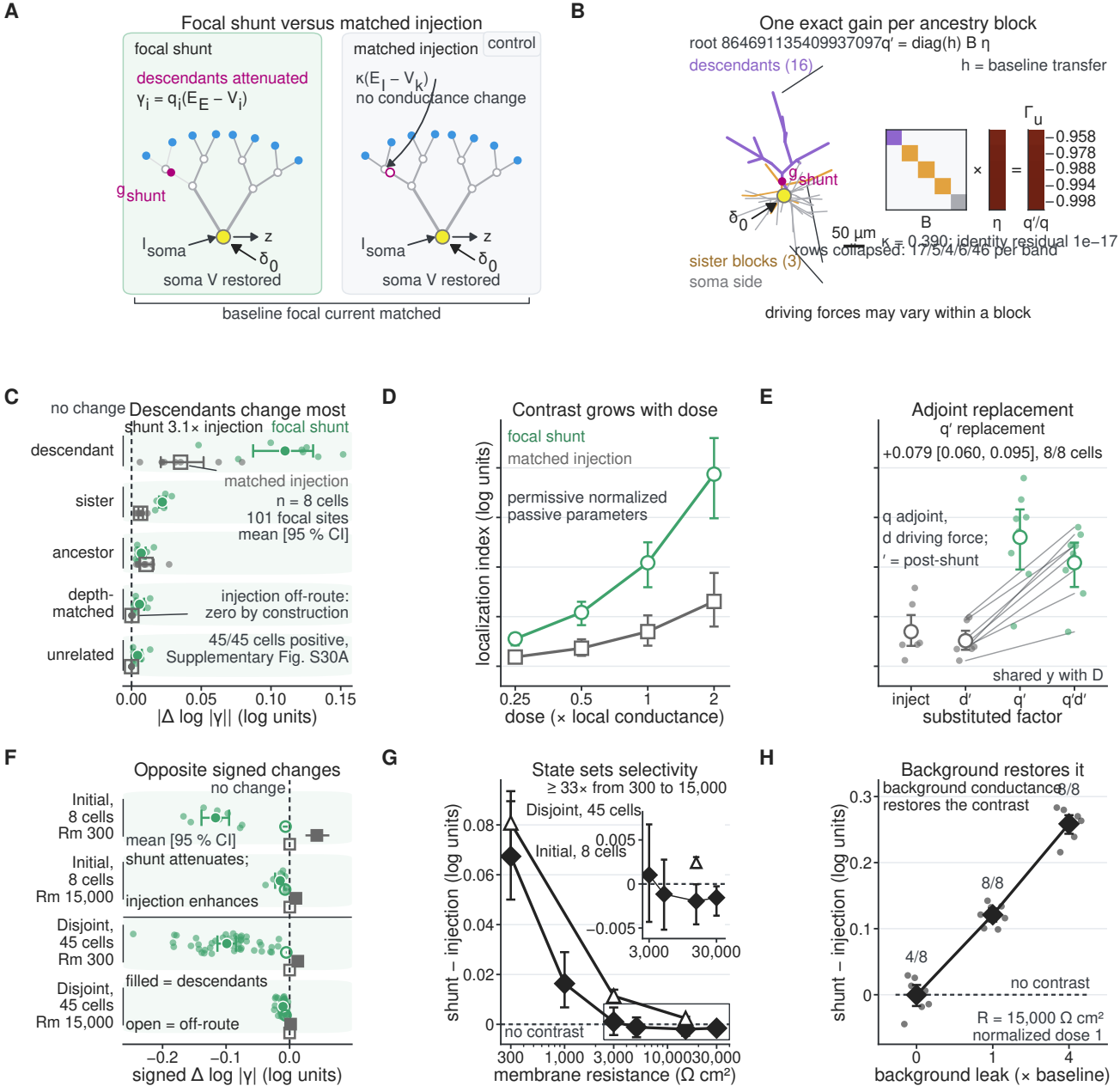

A
Focal shunt versus matched injection
focal shunt
descendants attenuated
$\gamma_i = q_i(E_E - V_i)$
g_{shunt}
I_{soma}
z
δ_0
soma V restored
matched injection
control
$\kappa(E_l - V_k)$
no conductance change
I_{soma}
z
δ_0
soma V restored
baseline focal current matched
B
One exact gain per ancestry block
root 864691135409937097q′ = diag(h) B η
descendants (16)
h = baseline transfer
g_{shunt}
δ_0
50 µm
sister blocks (3)
soma side
B
η
Γ_u
q′/q
0.958
0.978
0.988
0.994
0.998
κ = 0.390; identity residual 1e−17
rows collapsed: 17/5/4/6/46 per band
driving forces may vary within a block
C
Descendants change most
no change
shunt 3.1× injection
focal shunt
descendant
matched injection
sister
n = 8 cells
101 focal sites
mean [95 % CI]
ancestor
depth-matched
injection off-route:
zero by construction
unrelated
45/45 cells positive,
Supplementary Fig. S30A
0.00
0.05
0.10
0.15
|Δ log |γ|| (log units)
D
Contrast grows with dose
focal shunt
matched injection
permissive normalized
passive parameters
localization index (log units)
0.25
0.5
1
2
dose (× local conductance)
E
Adjoint replacement
q′ replacement
+0.079 [0.060, 0.095], 8/8 cells
q adjoint,
d driving force;
′ = post-shunt
shared y with D
inject
d′
q′
q′d′
substituted factor
F
Opposite signed changes
no change
Initial, 8 cells Rm 300
mean [95 % CI]
Initial, 8 cells Rm 15,000
shunt attenuates;
injection enhances
Disjoint, 45 cells Rm 300
Disjoint, 45 cells Rm 15,000
filled = descendants
open = off-route
−0.2
−0.1
0.0
signed Δ log |γ| (log units)
G
State sets selectivity
≥ 33× from 300 to 15,000
Disjoint, 45 cells
Initial, 8 cells
0.005
0
−0.005
3,000
30,000
shunt − injection (log units)
0.08
0.06
0.04
0.02
0
no contrast
300
1,000
3,000
10,000
30,000
membrane resistance (Ω cm²)
H
Background restores it
background conductance restores the contrast
8/8
8/8
4/8
no contrast
R = 15,000 Ω cm²
normalized dose 1
shunt − injection (log units)
0.3
0.2
0.1
0
0
1
4
background leak (× baseline)